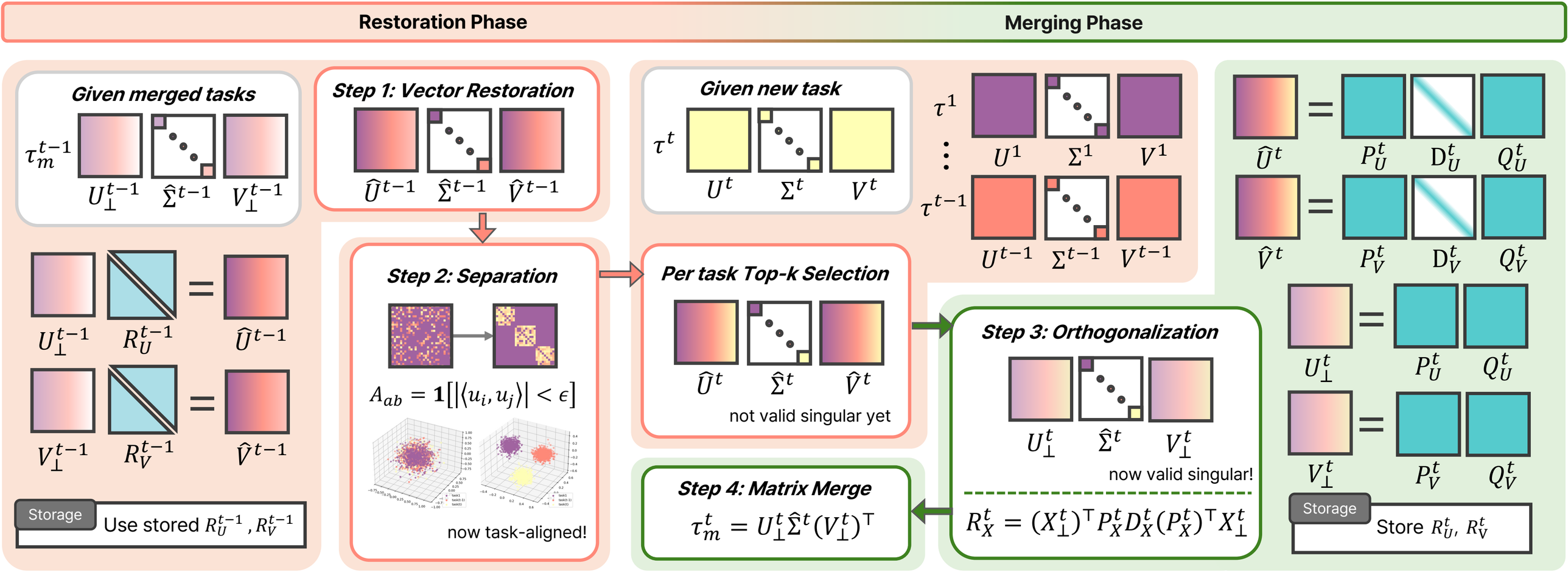

Restoration Phase
Merging Phase
Given merged tasks
τ_m^{t-1}
$U_\perp^{t-1}$ $\hat{\Sigma}^{t-1}$ $V_\perp^{t-1}$
$U_\perp^{t-1}$ R_U^{t-1} = $\hat{U}^{t-1}$
$V_\perp^{t-1}$ R_V^{t-1} = $\hat{V}^{t-1}$
Storage
Use stored R_U^{t-1} , R_V^{t-1}
Step 1: Vector Restoration
$\hat{U}^{t-1}$ $\hat{\Sigma}^{t-1}$ $\hat{V}^{t-1}$
Step 2: Separation
$A_{ab} = \mathbf{1}[|\langle u_i, u_j \rangle| < \epsilon]$
now task-aligned!
Given new task
τ^t
U^t Σ^t V^t
Per task Top-k Selection
$\hat{U}^t$ $\hat{\Sigma}^t$ $\hat{V}^t$
not valid singular yet
τ^1 U^1 Σ^1 V^1
τ^{t-1} U^{t-1} Σ^{t-1} V^{t-1}
Step 3: Orthogonalization
$U_\perp^t$ $\hat{\Sigma}^t$ $V_\perp^t$
now valid singular!
$R_X^t = (X_\perp^t)^\top P_X^t D_X^t (P_X^t)^\top X_\perp^t$
Step 4: Matrix Merge
$\tau_m^t = U_\perp^t \hat{\Sigma}^t (V_\perp^t)^\top$
$\hat{U}^t$ = P_U^t D_U^t Q_U^t
$\hat{V}^t$ = P_V^t D_V^t Q_V^t
$U_\perp^t$ = P_U^t Q_U^t
$V_\perp^t$ = P_V^t Q_V^t
Storage
Store R_U^t, R_V^t